

Next, configure the Android SDK. Begin by downloading the SDK binary for Linux from the Android Developer website (the link is in Reference). Extract the contents of the TGZ archive using any archive extraction tool. Note the location you extract it from—you will need the path later on.

You then need to install the Android Developer Tools (ADT) plug-in for Eclipse. Start Eclipse, and go to *Help > Install New Software*. Click the 'Add' button to add a repository. Enter a name of your choice for the repo, and for Location, enter <https://dl-ssl.google.com/android/eclipse/>.

Clicking 'OK' will bring up a checklist. Check the 'Developer Tools' item and click *Next* to view a list of what it is going to download. Proceed, and accept the *License Agreement* on the next screen. Hit the 'Finish' button, and wait during the download and install. Once the ADT plug-in is installed and Eclipse has been restarted, go to *Window > Preferences > Android* and set the SDK path to the path of the extracted SDK. Next, go to *Window > Android SDK and AVD Manager*. Select the *Available Packages* category, and check all items to do a full install. Click 'Install Selected' to proceed. The install will take some time, mostly for downloads. Grab a cup of coffee or your favourite beverage while you wait.

## Simple app

With the environment for Android development set up, let's make a HTML file and then use PhoneGap to build the app. Open up *gedit* and enter the following HTML, saving it as "index.html":

```
<html>
<head>
<script type="text/javascript" charset="utf-8"
src="phonegap-0.9.6.js"></script>
</head>
<body>
<h1 align="center">Hello World!</h1>
<h2 align="center">from PhoneGap</h2>
</body>
</html>
```

Before creating a new project in Eclipse, you need to create an Android Virtual Device (AVD) in which to test the app before installing it to your phone. In Eclipse, open *Window > Android SDK and AVD Manager*. Select 'Virtual Devices' and then 'New'. In the dialogue box, give a device name and select a platform (say, Android 2.3). Click 'Create AVD' to finish.

Now for a new Eclipse project. In Eclipse, go to *File > New > Project*. For Project type, select Android Project. Fill in the following details:

- **Project Name:** 'HelloWorld'.
- **Application Name:** "My First Android App" (or your choice)

- **Package Name:** the name-space of the app—what we can loosely call the 'context' of the app in development terms. Use 'com.phonegap.helloworld'; phonegap has to be a part of the name-space since we will be wrapping the app with it.
- **Create Activity:** Keep this as *App* for all PhoneGap-built apps.
- **Build Target:** Select 'Android 2.3'.  
Click 'Finish'; and Eclipse prepares the project structure and files.

## Integrating PhoneGap into the project



**Note:** Some of these steps involve entering code that is repeated for all apps using PhoneGap. The PhoneGap site has this code ready to copy and paste, to save effort. See the link in *References* for this code.

First, download the PhoneGap binary from the PhoneGap website and extract it. In Eclipse, create two folders in the root of the project: "/assets/www" and "/libs". Place the HTML file we created into 'www'. In the HTML code, replace the name (depending on the version and filename) of the PhoneGap JS file you are going to use. From the *Android* folder of the extracted PhoneGap archive, copy the JS file to the Eclipse project's 'www' folder, and

## Go Online to learn Linux Device Drivers



Core skills like kernel programming or Linux device drivers should be learned at a slow and steady pace for fully comprehending it, says Mr. Raghu Bharadwaj, a lead trainer on core Linux programming. Unlike normal application software learning where it is mostly restricted to functionality based learning, system programming involves deep understanding and analysis of issues like the Linux kernel and its subsystems. Your options to learn these skills are classroom based training, learning through experience, self learning through trial and error and through an effective online course.

While all other modes of training has its own merits and demerits, online learning is definitely a step ahead, especially for courses like this. Online courses gives you the comfort of accessing it 24x7 and lets you learn things during hours of focus and interest. Another important aspect is that it permits you to go through each session any number of times and fully understand each and every topic effectively.

Mr. Raghu Bharadwaj is a lead trainer at Veda Solutions, you can listen to his **training videos** on Linux kernel and device drivers at [www.techveda.org](http://www.techveda.org)

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